ANDREW ZENG

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EDUCATION

Northeastern University **Boston, MA**
B.S. in Electrical Engineering, Minor in Computer Science *May 2028*
GPA: 3.29
Honors: Northeastern University Merit Scholarship, Global Scholars program
Activities: NU Lunabotics, Google Developer Group, NEU Club Golf, Husky Trek
Coursework: Embedded Design, Computing Fundamentals for Engineers, Fundamentals of CS II, Discrete Structures

SKILLS

Technical: Python, MATLAB, JavaScript, C++, AutoCAD
Hardware: Arduino, Raspberry Pi, PCB design, soldering, laser cutting, rapid prototyping, 3D Printing
Tools & Frameworks: Manim, SolidWorks, Jupyter Notebook, VSCode, MacOS, Windows, KICAD

EXPERIENCE

University of Michigan **Ann Arbor, MI**
Research Intern *May 2025 – Present*
- Integrate sensor data with Python machine learning AI systems on custom PCBs for autonomous driving vision
- Develop Python-Manim visualizations for university lectures on core automotive electrical engineering topics
- Integrate advanced perception and control systems in RC car platforms illustrating autonomous driving concepts

Google Developer Group **Oakland, CA**
Lead Partnerships Coordinator *August 2024 – May 2025*
- Initiated and maintained relationships with 50+ tech companies and partners to foster collaboration and growth
- Curated and executed meetings, development sessions, and networking events for over 500 freshmen on campus
- Led a team of partnership coordinators to maintain partner engagement, and streamline collaboration

PERSONAL PROJECTS AND PUBLICATIONS

Custom Bus PCB *January 2026 - Present*
- Fabricate custom bus PCB using I^2C communication system to coordinate sensor boards in rover system
- Create schematics in KiCAD, including signal routing, connectors, and interface logic for board communication
- Implement power management architecture, distributing power to sensor components ensuring correct output

Self-Driving RC Car *May 2024 - August 2024*
- Designed in AutoCAD and fabricated 3D-printed and laser-cut car utilizing Arduino Nano hardware and sensors
- Achieved autonomous navigation through Python-trained ML models with live ultrasonic and IR sensor feedback
- Developed vehicle models using Python, integrating Arduino-based feedback and AutoCAD chassis components

Electric Go-Kart *March 2024 - December 2024*
- Modeled and fabricated custom go-kart components using SolidWorks with FEA-based load optimization
- Installed, configured, and soldered motor controller and power distribution systems within chassis for operation
- Designed and implemented electrical wiring architecture to integrate motor control and high-voltage battery

"DEMONSTRATION OF THE HYDROGEN CYCLE USING A PEM FUEL CELL"
by Andrew Zeng, Devon Renock- PhD *May 2023 - August 2023*
- Researched Fuel Cells and increased efficiency by 15% using membranes containing higher proton conductivity
- Published findings as the only high school presenter at the 2023 American Chemical Society Fall Conference
- Granted sponsor award from USAF and Point Park University Scholarship worth ~$30,000 yearly at PRSEF 2023